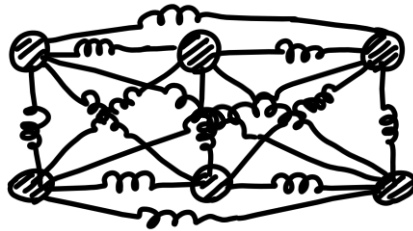
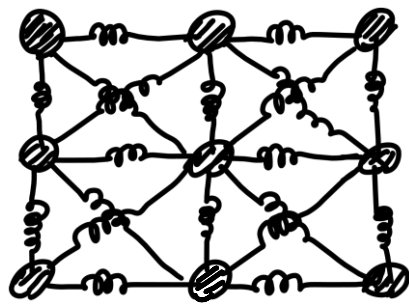


- مروری بر مدل بولتزمن.
- مدل دبی.
- مدل اینشتین در آتزانبل ماکرونیک.
- مدل اینشتین در آتزانبل میکرو ماکرونیک.



- مروری بر مدل بولتزمن:

$$H = \frac{P^T P}{2m} + \frac{1}{2} X^T K X \rightarrow H = \frac{\tilde{P}^T P}{2m} + \frac{1}{2} \tilde{X}^T \tilde{K} \tilde{X}$$

$$H = \frac{1}{2m} (P_1^2 + P_2^2 + \dots + P_{3N}^2) + \frac{1}{2} m (\omega_1^2 Q_1^2 + \omega_2^2 Q_2^2 + \dots + \omega_{3N}^2 Q_{3N}^2)$$

$$\rightarrow \langle E \rangle = 3Nk_B T \Rightarrow C_v = \frac{\partial \langle E \rangle}{\partial T} = 3Nk_B$$

$$\hat{H} = \sum_{i=1}^{3N} \frac{\hat{p}_i^2}{2m} + \frac{1}{2} \omega_i^2 \hat{Q}_i^2 \Rightarrow \hat{H} |n_1, \dots, n_{3N}\rangle = E_{\vec{n}} |n_1, \dots, n_{3N}\rangle$$

$$E_{\vec{n}} = \sum_{i=1}^{3N} \hbar \omega_i (n_i + \frac{1}{2}) \quad n_i = 1, 2, \dots$$

$$E_n = \hbar \omega (n + \frac{1}{2})$$

مک فوسالکر کو انتوی یک بعدی:

$$Z_{1D} = \sum_{n=0}^{\infty} e^{-\beta \hbar \omega (n + \frac{1}{2})} = e^{-\beta \hbar \omega / 2} \sum_{n=0}^{\infty} e^{-n \beta \hbar \omega} = \frac{e^{-\beta \hbar \omega / 2}}{1 - e^{-\beta \hbar \omega}}$$

$$= \frac{1 \times 2}{2(e^{\beta \hbar \omega / 2} - e^{-\beta \hbar \omega / 2})} = \frac{1}{2 \sinh(\beta \hbar \omega / 2)}$$

$$\ln Z_D = -\ln 2 - \ln \left(\sinh \left(\beta \hbar \omega / 2 \right) \right)$$

$$\langle E \rangle_D = -\frac{\partial}{\partial \beta} \ln Z_D = \frac{\hbar \omega}{2} \coth \left(\beta \hbar \omega / 2 \right) = \frac{\hbar \omega}{2} \frac{e^{\beta \hbar \omega / 2} + e^{-\beta \hbar \omega / 2}}{e^{\beta \hbar \omega / 2} - e^{-\beta \hbar \omega / 2}}$$

$$= \frac{\hbar \omega}{2} \frac{e^{\beta \hbar \omega} + 1}{e^{\beta \hbar \omega} - 1} = \hbar \omega \left(n_B(\hbar \omega \beta) + \frac{1}{2} \right) \quad \underline{n_B(n) := \frac{1}{e^n - 1}}$$

$$\frac{1}{e^{\beta \hbar \omega} - 1} + \frac{1}{2} = \frac{1}{2} \frac{e^{\beta \hbar \omega} + 1}{e^{\beta \hbar \omega} - 1}$$

$$\sum_{n=1}^{\infty} e^{-n\eta} \quad \uparrow$$

$$\langle E \rangle_D = \hbar \omega \left(n_B(\beta \hbar \omega) + \frac{1}{2} \right)$$

$$\beta = \frac{1}{kT} = -\frac{1}{kT^2} = -k\beta^2$$

$$C = \frac{\partial}{\partial T} \langle E \rangle_D = \left(\frac{\partial}{\partial \beta} \langle E \rangle_D \right) \left(\frac{\partial \beta}{\partial T} \right) = -k\beta^2 \frac{\partial}{\partial \beta} \langle E \rangle_D$$

$$U = \frac{\hbar \omega}{2} \frac{e^{\beta \hbar \omega} + 1}{e^{\beta \hbar \omega} - 1} \Rightarrow C = -\frac{\hbar^2 \omega^2}{2} k\beta^2 \frac{e^{\beta \hbar \omega} (e^{\beta \hbar \omega} - 1) - e^{\beta \hbar \omega} (e^{\beta \hbar \omega} + 1)}{(e^{\beta \hbar \omega} - 1)^2}$$

$$= C = k (\beta \hbar \omega)^2 \frac{e^{\beta \hbar \omega}}{(e^{\beta \hbar \omega} - 1)^2}$$

$$C \approx k_B (\beta \hbar \omega)^2 \frac{1}{(\beta \hbar \omega)^2} = \underline{k_B} \quad : \beta \hbar \omega = \frac{\hbar \omega}{kT} \rightarrow 0 \quad (1)$$

$$C \approx k (\beta \hbar \omega)^2 \frac{1}{e^{\beta \hbar \omega}} \rightarrow 0 \quad : \beta \hbar \omega \rightarrow \infty \quad (2)$$

$$E_{n_x, n_y, n_z} = \hbar \omega \left[(n_x + \frac{1}{2}) + (n_y + \frac{1}{2}) + (n_z + \frac{1}{2}) \right] \quad : \text{w. w. } \underline{\text{w. w.}}$$

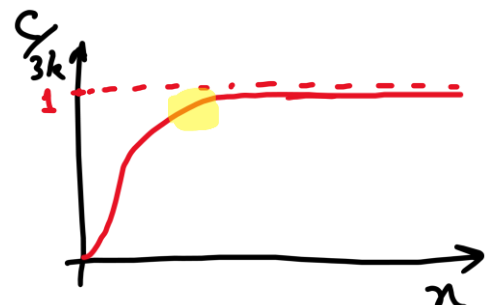
$$Z_{3D} = \sum_{n_x, n_y, n_z} e^{-\beta E_{n_x, n_y, n_z}} = (Z_{1D})^3 \Rightarrow \underline{Z_{3D} = 3 \ln Z_{1D}}$$

$$\Rightarrow \langle E \rangle_{3D} = 3 \langle E \rangle_{1D} \Rightarrow C_{3D} = 3 C_{1D}$$

$$C_{\text{per atom}} = 3 k_B (\beta \hbar \omega)^2 \frac{e^{\beta \hbar \omega}}{(e^{\beta \hbar \omega} - 1)^2}$$

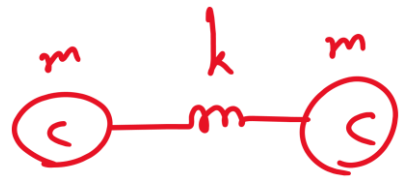
$$C_{3k} = (\beta \hbar \omega)^2 \frac{e^{\beta \hbar \omega}}{(e^{\beta \hbar \omega} - 1)^2} \xrightarrow[n = \frac{kT}{\hbar \omega}]{n = \frac{1}{\beta \hbar \omega}} C_{3k} = \frac{1}{n^2} \frac{e^{1/n}}{(e^{1/n} - 1)^2}$$

$$\hbar \omega_{\text{Einstein}} = \underline{k_B T_E}$$

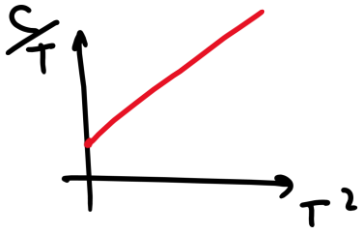
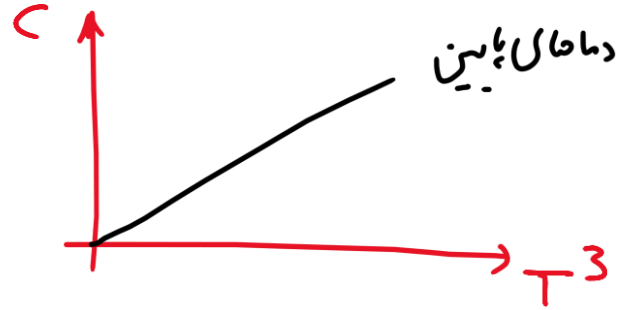


$$\omega_E \propto \sqrt{\frac{k}{m}}$$

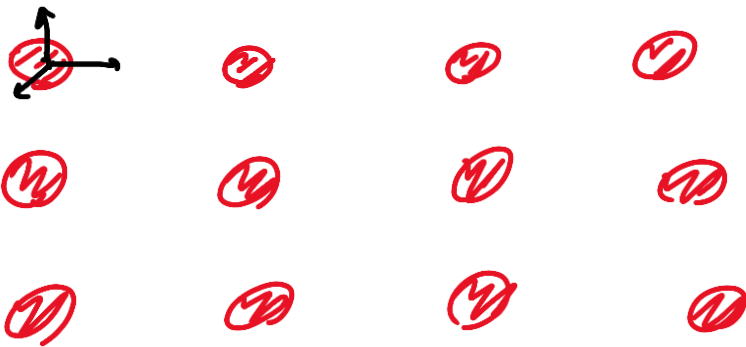
$$\omega_E^e > \omega_E^a$$



$$T_E = \frac{\hbar \omega_E}{k_B}$$



برای دماهای غیرمتناهی: $C \propto T^3 + aT$



مدل دبی:

$$\langle E \rangle = 3 \hbar \omega_{(\vec{k})} \left(n_B(\hbar \omega_{(\vec{k})}) + \frac{1}{2} \right)$$

$$\omega = v |\vec{k}|$$

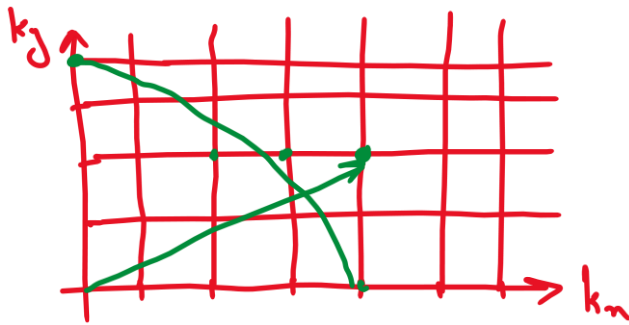
$$\langle E \rangle = 3 \sum_{\vec{k}} \hbar \omega_{(\vec{k})} \left(n_B(\beta \hbar \omega_{(\vec{k})}) + \frac{1}{2} \right)$$

$$\psi = A \sin k_n n + B e^{ik_n n}$$

$$n = 1, 2, \dots$$

$$\psi = A \sin k_n x \xrightarrow{n=L} 0 = \sin k_n L \Rightarrow k_n L = n\pi$$

$$0 < k_n, k_y, k_z \Rightarrow k_n = \frac{\pi}{L} n_n, \quad k_y = \frac{\pi}{L} n_y, \quad k_z = \frac{\pi}{L} n_z$$



$$\Delta k_n \Delta k_y \Delta k_z = \frac{\pi^3}{L^3} \Delta n_n \Delta n_y \Delta n_z$$

$$dn = \frac{L^3}{\pi^3} \frac{1}{8} dk_n dk_y dk_z = \frac{L^3}{(2\pi)^3} d\vec{k}$$

$$\sum_{\vec{k}} \longrightarrow \frac{L^3}{(2\pi)^3} \int d\vec{k}$$

$$\langle E \rangle = 3 \frac{L^3}{(2\pi)^3} \int d\vec{k} (\hbar \omega_{(\vec{k})}) \left(n_B(\beta \hbar \omega_{(\vec{k})}) + \frac{1}{2} \right)$$

$$\langle E \rangle = 3 \frac{4\pi L^3}{(2\pi)^3} \int_0^\infty k^2 dk (\hbar \omega_{(k)}) \left(n_B(\beta \hbar \omega_{(k)}) + \frac{1}{2} \right)$$

$$k v = \omega_{(k)} \Rightarrow \langle E \rangle = \frac{3L^3}{2\pi^2} \int_0^\infty \omega^2 d\omega \left(\frac{1}{v^3} \right) (\hbar \omega) \left(n_B(\beta \hbar \omega) + \frac{1}{2} \right)$$

$$= \frac{3L^3 \hbar}{2\pi^2 v^3} \int_0^\infty \omega^3 d\omega \left(\frac{1}{e^{\beta \hbar \omega} - 1} + \frac{1}{2} \right)$$

دائری به T ندارد: بی نهایت است.

$$= \frac{3l^3 \hbar}{2\pi^2 v^3} \int_0^\infty \omega^3 d\omega \frac{1}{e^{\beta \hbar \omega} - 1} + A$$

$$\beta \hbar \omega = n$$

$$= \underbrace{\frac{3l^3 \hbar}{2\pi^2 v^3} \frac{1}{(\beta \hbar)^4}}_B \int_0^\infty n^3 dn \frac{1}{e^n - 1} = B \int_0^\infty n^3 \frac{e^{-n}}{1 - e^{-n}} dn$$

$$= B \int_0^\infty n^3 \sum_{n=1}^\infty e^{-nn} dn \xrightarrow{nn=y} B \sum_{n=1}^\infty \frac{1}{n^4} \int_0^\infty y^3 dy e^{-y}$$

$\Gamma(4) = 3!$

$\zeta(4) = \frac{\pi^4}{90}$

$$\Rightarrow B \frac{\pi^4}{90} \times 6 = B \frac{\pi^4}{15}$$

$$\langle E \rangle = \frac{l^3 \hbar k_B T^4}{2\pi^2 v^3 (\hbar)^4} \frac{\pi^4}{15} + A$$

$\frac{N}{L^3} := n$

$$\frac{N 9 l^3 \hbar}{N 6 \pi^2 v^3} = N \frac{9 \hbar}{6 \pi^2 n v^3} = N \frac{9 \hbar}{\omega_d^3}$$

$$\boxed{\omega_d^3 := 6 \pi^2 n v^3}$$

$$C = \frac{\partial}{\partial T} \langle E \rangle = \underline{\underline{D T^3}}$$

$$\int_0^{\omega_{\text{cutoff}}} g(\omega) d\omega = 3N$$

$$\langle E \rangle = 3 \frac{4\pi L^3}{(2\pi)^3} \int_0^{\omega_{\text{cutoff}}} k^2 dk (\hbar \omega_{(k)}) \left(n_B(\beta \hbar \omega_{(k)}) + \frac{1}{2} \right)$$